

It is given that the shortest distance between the lines l_1 and l_2 is $\sqrt{21}$.

- (a) Find the value of t . [5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

The plane Π_1 contains l_1 and is parallel to l_2 .

- (b) Write down an equation of Π_1 , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$. [1]

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The plane Π_2 has Cartesian equation $5x - 6y + 7z = 0$.

- (c) Find the acute angle between l_2 and Π_2 . [3]

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- (d) Find the acute angle between Π_1 and Π_2 . [3]

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